Project and Professionalism (6CS007)

Project Proposal

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1. Introduction

Today’s job seekers have more difficulty than ever before creating an impressive resume that reflects modern industry standards when applying for jobs that utilize an ATS (Applicant Tracking System). Most job seekers continue to create their resumes using traditional methods, such as applying via a template or manually formatting their resume, which not only results in a poorly crafted resume but also does not highlight relevant skills or experiences and fails to correlate to a specific job description that the candidate is applying for. Because of the increase in the use of automated screening tools by employers, the need for smarter AI generated resumes has dramatically increased.

The project provides an AI Powered Resume Generator, which uses Artificial Intelligence and Natural Language Processing, and is an online system that supports candidates by providing an efficient way to create resumes that are ATS compliant, receive immediate resume optimization recommendations, and have a resume tailored to the job descriptions submitted by employers. The system will also include cover letter generation, a career analysis tool, and mock interview training.

As we see more and more reliance on Artificial Intelligence in hiring practices, there is a growing need for tools to assess what job criteria are being used to evaluate a candidate's qualifications based on the information provided in the applicant's resume, to help applicants tailor the specifics of their resumes to match those criteria, and to assist applicants in identifying which jobs may be the best fit for them. This initiative addresses the failure of traditional resume development processes and provides applicants with a more informative, data driven tool for navigating their career advancement opportunities.

1.1. Project Domain

The AI Powered Resume Generator falls under because it covers the fields of Artificial Intelligence (AI), Machine Learning (ML), and Web Application Development (WAD). With the use of this platform for recruitment support, job seekers can produce high quality resumes that meet the requirements of an applicant tracking system (ATS), and access a variety of tools for increasing their careers via a web based interface. The system is an intuitive and automated tool that will assist with:

AI/NLP Document Processing; Resume Optimization Job Matching: Cloud Based Web Applications; Career Insights Automated Content Creation.

The overall goal of the AI Powered Resume Generator is to provide job seekers with the opportunity to use intelligent technology and the latest advancements in recruitment technology to produce a high quality resume, reflecting the standards of today's job marketplace.

1.2. Relevance

To address key challenges for both job seekers and employers in recruiting and reviewing resumes, this project addresses many significant problems faced by both job seekers and employers today:

- The majority of job seekers don't know how to write their resumes according to ATS specifications so their resumes could get disqualified without having been reviewed by a recruiter or employer.
- Most online resume building tools don't let users customize their resumes or allow them to create resumes specifically for the positions they are targeting.
- Creating a resume manually is time-consuming, and typically results in mistakes and/or improper presentation of information on a candidate's resume; thereby not maximizing the opportunity to highlight skills and experience most relevant to the job posting.
- The majority of job seekers do not have access to modern intelligent technology that provides them with immediate feedback and enables them to optimize their resume

content and provides insights on new career opportunities based on changing market trends and conditions.

This project is relevant because it introduces an AI powered solution that improves resume quality, increases visibility in digital hiring systems, and supports users with smart career guidance.

1.3. Motivation

The primary motivator behind this project came from recognizing that many prospective employees have been struggling to navigate a complicated modern-day recruitment atmosphere. Below are some of the main issues which inspired the creation of this project:

- The time and effort expended on poorly completed resumes. The vast majority of applicants will commit hours towards creating or editing their resumes only to have to go through several iterations before meeting the ATS (applicant tracking system) requirements; thus having fewer chances to be noticed by prospective employers.
- Lack of Real-Time Personalization/Feedback on Resumes. Traditional resume creation sites do not provide real-time suggestions to improve upon the applicant's resume and do not give insight on how to best promote their qualifications, skills, and experience related to their target employer.
- Disparities in Access to Career Development Tools. People living in rural areas or without established networks have little access to resume enhancement systems powered by AI that can help improve their resume quality and career opportunities.
- Ineffective Recruitment Methods. Employers receive a high volume of improperly formatted resumes, which creates a higher workload for human resource personnel and heats up the time frame for filling positions.

The goal of the current project is to rectify these issues by offering an AI-driven platform where users can build and improve their resumes more easily, get real-time optimization based on ATS and job posting requirements, and obtain career advice leading to increased employment opportunities.

2. Problem Statement

Many job seekers will face difficulties when creating an effective resume for one or more of the following reasons:

- The structure of a resume has to meet the standards set by Applicant Tracking Systems. If the resume is not formatted appropriately, it may be rejected before a reviewer even sees it.
- Updating and changing your resume for multiple applications takes a considerable amount of time, based on the applicant's individual levels of work experience.
- Job applicants generally have limited access to coaching or support that assist them in identifying their skill sets and experiences that relate to jobs.
- A challenge of matching resumes to job descriptions has resulted in reduced interview opportunities.
- There are no combined tools available that help with cover letter production, portfolio generation, or preparation for interviews.

Job seekers who encounter these challenges are unable to maximize their chances of earning interview invitations and expending effort searching for jobs, all of which contributes to increased stress throughout the course of their attempts to find work in a very competitive marketplace.

3. Project as a Solution

A Job Seeker's AI-Powered Resume Generator will provide several solutions to help job seekers overcome the problems they encounter while searching for jobs:

- An ATS Ready Resume Builder: A system for creating resumes that comply with the latest screening protocols in order to increase chances of being visible in hiring systems.
- An AI-Based Resume Optimizer: Offers "real time" feedback to improve resume content, structure, key terms and overall resume effectiveness.
- Job Description Evaluation: Uses Natural Language Processing (NLP) to evaluate the relationship between job descriptions and resumes to determine if all required skills/experience have been included, identify keywords that match job description and determine alignment between job description and resume.
- An AI Generated Cover Letter: Creates a new customized cover letter based on the user's experience and job requirements.
- The Career Development Insights Dashboard will provide users with Suggested Career Paths, Skills Needed for those Occupations, and Trends in the Field/Industry related to those suggested career paths.
- Mock Interview Preparation: The Mock Interview Preparation Tool is an AI-driven tool that allows users to prepare for Mock Interviews by providing tailored Mock Interview Questions for each user's desired job and offering feedback on how to improve their performance.
- Complete Portfolio Development Tool - A simple-to-use Complete Portfolio Development Tool to allow users to create a complete online portfolio showcasing their accomplishments and skills.

Using Artificial Intelligence, NLP and modern web technologies, the AI-Powered Resume Generator provides a comprehensive system for enhancing your career, helping you prepare for interviews, and ultimately improving your chances of getting hired.

4. Scope and Limitations

4.1. Project Scope

- **Core Capabilities:**
  - Registered users creating/editing their profiles
  - AI producing a resonating resume according to court and proofing of all documents via an ATS compliant system.
  - Real-time assessment and evaluation of resumes, with suggestions to improve.
  - A job description compared to a resume using Natural Language Processing technology.
  - AI will generate cover letters tailored to individual job postings.
  - The ability to build a portfolio and store information about the portfolio.
  - Mock-interviews through the generation of questions asked during the mock-interview.
  - Ability to download the resume and cover letter in multiple formats.

- **Technical Implementation:**
  - React for the front-end development of the web-application.
  - Python with either Flask or Django for development of backend services.
  - REST-based Architecture will be used for inter-module communication.
  - Creation and maintenance of a Database with either MongoDB or PostgreSQL.
  - Creation and incorporation of AI and NLP models used to analyze and generate text.
  - Secure authentication and protection of use data.

- **Language Support:**
  - English Language user interface.
  - Natural Language Processing (NLP) capabilities for extracting, enhancing and improving the quality, readability and usefulness of text.

- **Additional Enhancements (if time permits):**
  - Integrating future models of AI for a deeper level of personalization.
  - API-based job listing integration from many other service providers.
  - Integration of multiple languages for the creation of a resume.
  - Advanced analytic tool for predicting which type of area/position a person may be well-suited for in their career.

4.2. Limitations:

- The system should be able to match jobs to people accurately, however, it does not guarantee 100 percent accuracy due to both the types of data that are available and the reliance on AI predictions.
- The system will not validate the user's information through extensive background checks (if they provide information about prior jobs or experience).
- The project will only be operating as a prototype with limited template capability, limited AI functionality, and limited processing power.
- The prototype will not be able to work directly with third-party job boards to begin because the API does not offer access for use at the launch of the product.
- Advanced multilingual resume creation capabilities will not be available at the first release of the system.

5. FDD (Functional Decomposition Diagram)

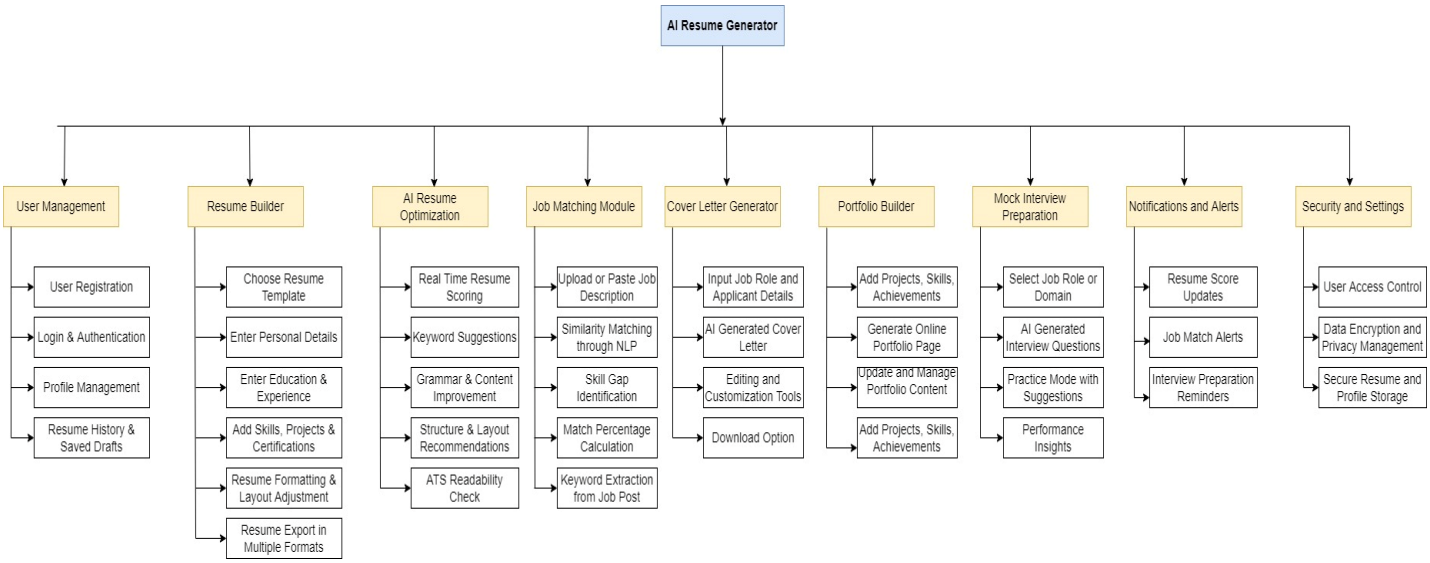

Figure 1 FDD (Functional Decomposition Diagram)

The Function Decomposition Diagram illustrates how the entire system of an AI Resume Builder is broken down into individual components. Modules of the System include User Management, Resume Builder, AI Resume Optimizer, Job Matching, Cover Letter Generator, Portfolio Builder, Mock Interview Prep, Notifications, and Security. Each Module is further decomposed into activities. Examples are registering users, inputting information, formatting resumes, scoring resumes, identifying skill gaps, comparing jobs to resumes, generating cover letters automatically, creating Portfolio pages, providing questions for mock interviews and securing user Profiles. The decomposition of the system clearly illustrates how each of these components works together to complete the overall flow of the AI Resume Builder.

6. Product Design

6.1. User Types

a) Job Seeker

Creates an account or logs in via a secure method, provides career information, uploads an old PDF resume, chooses a resume template, creates a resume via AI, changes and downloads final resume, and maintains a history of past resumes created via the system.

b) Recruiter

Creates an account or logs in, reviews candidate resumes that candidates have authorized for review, narrows down candidate resumes based on skill or experience, provides optional recommendations on how to improve resumes, and maintains a list of saved candidates' profiles.

c) Administrator

Maintains users of the system, monitors resume generation activity, verifies the accuracy of the AI model, updates the template library, and provides analytics on the system's performance.

6.2 High-Level User Workflows

6.2.1 Job Seeker Workflow

a) Registration or Login – through Secure Authentication – using JWT Token
b) Creation of a Profile – Basic Personal Information and Career History
c) Upload of Previous Resume – PDF Format if applicable.
    - Store PDF File – Protected Using AES 256 Encryption
    - Decrypt the uploaded file within memory to extract text for use.
d) Perform OCR using 'Tesseract' to Extract Text from Uploaded Document
e) Perform NLP on the uploaded document to identify Skills, Experience & Education, and Achievements using SpaCy and T5 small
f) Create Resume Content via AI Generation
    - Select a template
    - Structure Resume Content Automatically Created by AI
g) Preview Resume

- Modify and Personalize Tone, Wording, and Layout

h) Download Options:
    - PDF
    - DOCX

i) Preserve Resume History Securely for Future Editing

6.2.2. AI Resume Generator Flow

1. When Users create an account or log in to the platform to use it.
2. View Resume Generation → Access to AI-Generated Resumes and Skill Match Scores with Optimization Suggestions
3. Review Resume → Access AI Suggestions for Improvement by reviewing resume (format, content) with additional AI suggestions.
4. Edit Resume → An optional step to manually edit any sections of their resume that are unclear or critical based on the AI's recommendations.
5. Export and Keep Track of Your Resume / Download→ Once Users make all of the edits and finalize the export, they can send all of their exported resumes as PDFs/Word to keep track of which versions were sent to which companies so they can follow up if necessary.

7. Artifacts- Explanation of Subsystems

- **Resume Builder:** Produces resumes based on user information that meets ATS compliance.
- **Job Matching Module:** Matches resumes to job postings in real-time via NLP tech.
- **Scoring Module:** Provides feedback on how to optimize a resume.
- **Cover Letter Generator:** Generates a customized cover letter.
- **Portfolio Builder**: Users can then showcase projects or skills.
- **Mock Interview:** Users can then practice Q&A style as a preparatory tool.

8. Similar Systems

1. **<u>Canva Resume Builder</u>**

   Canva Resume Builder is an online tool designed to help a user create visually attractive resumes with pre-made templates. There are customization options for design, but the platform does not offer features for ATS optimization or AI matching based on jobs.

   - Type: Resume Design Tool
   - Focus: Creative templates and design freedom for resumes.
   - Access: Free and premium subscription for access; accessible online.
   - Best for: Users who want a visually attractive resume without heavy technical lifting.
   - Limitations: No ATS optimization or AI feedback.

2. **<u>Zety</u>**

   Zety is an online resume-building tool, with customizability and guided support to help you build your resume. Zety positions itself as user-friendly and quick, but it is not yet incorporating advanced AI elements into its service.

   - Type: Resume Builder Platform
   - Focus: Various customizable templates and basic resume building capabilities.
   - Access: Subscription based tool; available in most parts of the world.
   - Best for: For job seekers who are looking for simple, fast, and customizable resume options.
   - Limitations: Does not include features for AI-based job matching or resume scoring.

3. **Kickresume**

Kickresume is a resume and cover letter builder that enables users to create professional resumes in minutes. It does offer some basic portfolio options but does not have advanced AI-based optimization.

- Type: Resume & Cover Letter Builder
- Focus: The service provides templates, cover letter generation, and minimal portfolio functionality.
- Access: Free, with premium features; online.
- Best for: Anyone looking to build a resume who wants templates that come with cover letter support combined.
- Limitations: No ATS based optimization, no resume scoring or job-matching capabilities.

4. **Proposed System (AI Resume Generator)**

The AI-powered resume generator incorporates advanced AI & NLP capabilities to optimize resumes for ATS as well as optimize job matching precision. It also provides other career tools which other resume builders do not also offer.

- Type: AI-powered Resume and Career Support Platform
- Focus: ATS optimization, AI-powered job matching, resume scoring, generating cover letters, and career insights.
- Access: a web-based application; secure and accessible for all learners.
- Best for: job seekers, graduates or other working professionals who require an AI-supported tool to maximize hiring opportunities.
- Limitations: may require costs and/or technical resources in order to scale or incorporate API.

9. Comparison Table

Features	Canva Resume Builder	Zety	Kickresume	AI Resume Generator (Proposed)
Type	Resume Design Tool	Resume Builder Platform	Resume & Cover Letter Builder	AI-powered Resume & Career Support Platform
Focus	Creative templates and design freedom	Customizable templates, basic resume building	Templates, cover letter generation, minimal portfolio	ATS optimization, AI job matching, resume scoring, career insights
Access	Free & Premium, online	Subscription-based, widely accessible	Free & Premium, online	Web-based, secure, accessible for all learners
Best for	Users wanting visually attractive resumes	Job seekers wanting simple, fast, customizable resumes	Users needing resumes with cover letter support	Job seekers, graduates, professionals wanting AI-supported tools
ATS Optimization	✗	✗	✗	Available
AI Job Matching	✗	✗	✗	Available
Resume Scoring	✗	✗	✗	Available
Cover Letter Generation	✗	✗	Available	Available
Career Insights	✗	✗	✗	Available
Limitations	No ATS optimization or AI feedback	No AI-based job matching or resume scoring	No ATS optimization or job-matching	May require costs/technical resources to scale

Table 1 Comparison Table

10. Gantt Chart

A Gantt chart is a visual project management method that illustrates tasks, deadlines, and schedule over time. I am using it to plan, assign, and track the development tasks for my "AI Resume Generator" project to keep it organized and on track.

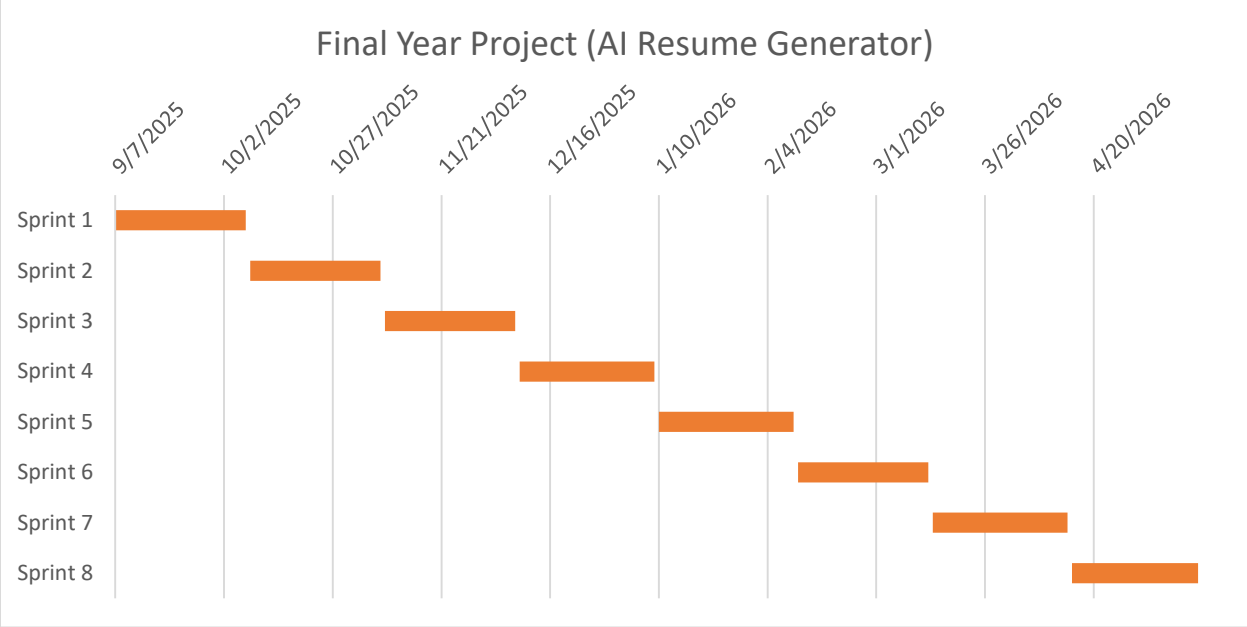

Figure 2 Gantt Chart

The Gantt Chart above shows the respective sprints for my AI Resume Generator project, indicating the timeline and order of the sprints, from September 07, 2025 to May 14, 2026. This tool is a valuable aid for project management and provides visualization of the development timeline, it can help track progress, and can determine whether all tasks/deliverables can be completed within the schedule.

11. Work Breakdown Structure (WBS)

The table below provides a breakdown of the tasks and work structure for all of the sprints of the AI Resume Generator project. Each sprint lasts approximately one month from 09/07/2025 to 05/14/2026.

No.	Sprint	Description	Start Date	End Date	Status
1	Sprint 1	Project Planning & Requirement Gathering	09/07/2025	10/07/2025	Done
2	Sprint 2	Design & Prototyping	10/08/2025	11/07/2025	Done
3	Sprint 3	Frontend + Backend Setup (Phase 1)	11/08/2025	12/08/2025	-
4	Sprint 4	Backend Expansion & Database Design	12/09/2025	01/09/2026	-
5	Sprint 5	Core Features Implementation	01/10/2026	02/10/2026	-
6	Sprint 6	Advanced Features & AI Integrations	02/11/2026	03/13/2026	-
7	Sprint 7	Testing & Bug Fixing	03/14/2026	04/14/2026	-
8	Sprint 8	Final Deployment & Documentation	04/15/2026	05/14/2026	-

Table 2 WBS

As mentioned above, this is the Work Breakdown Structure (WBS) for my AI Resume Generator project. Each sprint is about one month and has its own defined tasks. This will be important to allow me to manage my time and the tasks properly, while being able to monitor progress in a methodical way throughout the life of the project.

12. Chosen Software Development Methodology

An Agile (Scrum) methodology.

- Utilizing sprints (Design → Build → Test → Deploy).
- It is a perfect fit for a modular system (Resume Builder, Job Matching, Scoring, Cover Letter, Portfolio, Mock Interview).
- It is an iterative approach which gets feedback from testing and evaluation continuously.
- The timeline would be managed with a Gantt Chart and Work Breakdown Structure (WBS).

13. Chosen Tools and Technology

- **Frontend:** React.js
- **Backend:** Flask/Django + Node.js
- **AI/NLP:** GPT, spaCy, NLTK, BERT
- **Database:** MongoDB/PostgreSQL
- **Hosting:** AWS/Heroku
- **Version Control:** GitHub
- **Design:** Figma

14. References

There are no sources in the current document.